Romain Fillon

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French citizen (EU), born Dec. 1996

Fields: environmental and climate economics, macro dynamics
Methods: dynamic stochastic optimization, spatial models

Professional appointments

Institut Avant-garde, European Macro Policy Network Since Oct. 2026
Senior Economist. Developing heterogeneous-agent macro models of fiscal and climate policy, with a front-end interface for policymakers and citizens, in the context of France's 2027 presidential election.

The World Bank 2025–2026
Research consultant. Economics of flood risk management in World Bank risk reduction projects.

Education

PhD in Economics, Université Paris-Saclay (CIRED & PSAE) Dec. 2024
⋆ *Best Doctoral Dissertation in Environmental Economics in Europe* (EAERE, 2025)
⋆ *Best Doctoral Dissertation in Economics in Paris area* (Chancellerie de Paris, 2025)

Advisors: Céline Guivarch (CIRED) & Vincent Martinet (INRAE)
Committee: M. Fleurbaey (PSE), F. Henriet (AMSE), F. C. Moore (UC Davis), F. Venmans (LSE)
Fulbright Visiting Scholar, Columbia University, New York Sept. 2023 – Jan. 2024

MRes in Environmental Economics, Université Paris-Saclay 2021
ENS Paris-Saclay, Social Sciences, major in Economics 2015–2021
MSc in Management, ESSEC Business School 2020
BSc in Economics, Université Paris 1 Panthéon-Sorbonne 2016

Research

Job market paper

- 'The spatial climate risk premium: an application to Europe' [pdf upon request]
  I measure climate risk across European regions in a dynamic spatial model with adaptation.

Submitted

- 'Revisiting the global distribution of climate damages' [code, pdf]
  Cold economies are sometimes thought to gain from warming; with daily temperatures, I show they do not.
- 'Valuing climate subsystems: an application to the Amazon rainforest', with Guivarch [code, pdf]
  Modelling subsystems as dynamic assets rather than static stocks changes their value at global and local scales.
- 'Pricing risk at a boundary' [code, pdf]
  Tipping risk cannot be priced by its probability alone: policy needs a date of no return alongside the carbon price.
- 'Climate shift and economic damages', with Linsenmeier (Frankfurt) & Wagner (Columbia) [pdf]
  Beyond annual means, shifting daily temperatures raise global damages, unevenly across regions.

Working papers

- 'The spatial scale of climate damages' [pdf upon request]
  Urban heat falls below the resolution of climate-economy models; household sorting cannot offset what they miss.
- 'Who gains from flood protection?', with Avner & Hallegatte (WB) [pdf upon request]
  A framework to value World Bank flood risk reduction projects beyond expected annual damages and direct impacts.

Work in progress

- 'Distributional spatial integrated assessment models: an application to Europe'
- 'Is past natural climate variability a good proxy for future warming from carbon concentration?'

Publication

- Fillon, Guivarch & Taconet (2023). 'Optimal climate policy under tipping risk and temporal risk aversion', *Journal of Environmental Economics and Management* [code, coverage, pdf]

Awards and grants

2025 Best European Doctoral Dissertation in Environmental Economics, EAERE
2025 Best Doctoral Dissertation in Economics, First Prize, Chancellerie des Universités de Paris
2023 Fulbright grant, Columbia University (Sept. 2023 – Jan. 2024), plus Paris Region mobility grant
2021 3-year PhD grant, Université Paris-Saclay (2021–2024); 3-month extension, École des Ponts (2024)
2015 4-year fully funded studies, École Normale Supérieure Paris-Saclay (2015–2021)

Presentations

Invited seminars

2025 Bocconi University (Department of Economics, job talk)
2024 iRisk seminar (IÉSEG & LEM, Lille)
2023 MIT (CEEPR), Columbia (SIPA, Sust. Dev. Colloquium), Yale (Environment), Duke (UPEP)

Conferences

2026 WCERE (Nova SBE, Lisbon), SURED (Basel)
2025 EAERE (NHH, Bergen), RIEF (AMSE, Marseille)
2024 EAERE (KU Leuven), LAGV (AMSE, Marseille), FAERE (BETA, Strasbourg)
2022 EAERE (Bologna), PET (AMSE, Marseille), FAERE (Rouen), MUSEES (Lyon), FAERE Workshop

Other seminars

PSAE (×2, 2025), CIRED (2024), Paris-Saclay PhD seminars (2022–2024), Oxford CBT (2020)

Teaching and service

Teaching assistant – Econometrics (graduate, Sciences Po Paris, Prof. Junnan He, Fall 2022); Climate-economy models (graduate, Université Paris-Saclay, Prof. Céline Guivarch, Fall 2022)
Co-organizer – Parisian PhD Seminar on Economics and Environment (first edition 2024)
Referee – *Resource and Energy Economics*, *World Development*

Earlier positions

2019	Research Assistant, University of Oxford, with Prof. Eddy H. F. Tam	9 months
2019	Intern, UN World Food Programme, supply chain division, Rome	3 months
2017	Intern, *Cour des comptes* (French supreme audit institution), Paris	4 months
2016	Intern, Consulate General of France in Shanghai	3 months

Skills and languages

Programming: Python, R, Stata, LaTeX, Git/GitHub
Languages: French (native), English (fluent), German (intermediate)

References

Céline Guivarch, CIRED, École des Ponts (PhD advisor) [mail]
Vincent Martinet, PSAE, INRAE, Université Paris-Saclay (PhD advisor) [mail]
Stéphane Hallegatte, The World Bank [mail]
Gernot Wagner, Columbia Business School [mail]